INCH-POUND

MIL-W-63377 (AR)
AMENDMENT 2
1 February 1991
SUPERSEDING
AMENDMENT 1
18 MAY 1990

MILITARY SPECIFICATION

WARHEAD, ILLUMINATING, M257 W/M442 FUZE
(2.75 INCH ROCKET ILLUMINATING FLARE)

This amendment forms a part of Military Specification MIL-W-63377 (AR), dated 29 April 1987, and is approved for use within the U.S. Army Armament, Munitions, and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

3.4.1, Delete in its entirety and substitute:

"3.4.1, Function Time. The time between rocket motor ignition and flare candle ignition shall be 13.5 ± 2.0 seconds with Mk40 motor and 13.5 ± 2.5 seconds with Mk 66 motor."

PAGE 5

3.14, Delete "20 pounds is applied to the sear." and substitute, "less than 20 pounds is applied to the pull wire."

* Add the following paragraph:

"3.14.1 Nonfunction Drop Test. The gas generator shall not function when subjected to a seven foot drop to a steel plate five times in a pipe fixture."

3.17, Delete "50 pounds minimum and 75 pounds maximum" and substitute, "40 pounds minimum and 65 pounds maximum".

3.18, Delete "16 g's plus or minus .5 g's" and substitute, "12 to 14 g's".

3.19, Delete "twenty-two (22) g's plus or minus .5 g's" and substitute, "20 to 22 g's".

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PAGE 7

* 4.3.1, Delete "Gas Generator 9332277 30"
and substitute "Gas Generator 9332277 50"

PAGE 11

* Table I: Under "Gas Generator" Examination or Test:
Add the following:

"Nonfunction Drop Test 20 3.14.1 4.4.2.20".

PAGE 34

4.4.2.16, Delete Major 102 in its entirety.

PAGE 38

* 4.4.2.20, Add Critical 2 as follows:

"2 Nonfunction Drop Test 10 --- 3.14.1 4.5.7".

PAGE 47

4.5.3.1, Delete "4.3.4" and substitute "4.4.4".

PAGE 51

4.4.2.33, Delete Major 102 in its entirety.

PAGE 53

4.4.2.35, Delete Major 108 in its entirety.

PAGE 61

4.5.7, Delete "20 pounds on the sear." and substitute "less than 20 pounds on the pull wire.".

Delete "10 pounds on the sear." and substitute "less than 10 pounds on the pull wire."

Delete "sear movement or gas generator function." and substitiute "pull wire extraction or gas generator function."

* Add the following paragraph:

"4.5.7.1, Gas generator nonfunction drop test. For First Article, twenty (20) each gas generator shall be subject to a drop test, five (5) times each, installed loosely into a 0.50 in. dia. x 6.00 in. long steel pipe. The ends of the pipe shall be capped to prevent ejecting the gas generator during testing.

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The test fixture shall be dropped horizontal from a height of seven (7) feet min. to a steel plate. The First Article sample shall be rejected if one or more gas generator has functioned. For production, ten (10) each gas generators from each production lot shall be subjected to the drop test. If one or more gas generator has functioned, the production lot shall be rejected."

PAGE 62

4.5.10, Delete "16 g's plus or minus .50 g's" and substitute "12 to 14 g's".

Delete, in last sentence, "classed defective." and substitute "classed defective and removed from the lot."

4.5.11, Delete "22 g's plus or minus .50 g's" and substitute "20 to 22 g's".
Delete, in last sentence, "classed defective." and substitute "classed defective and removed from the lot."

4.5.12, Delete, in last sentence, "classed defective." and substitute "classed defective and removed from the lot."

4.5.13, Delete "A rocket motor shall be assembled" and substitute "A rocket motor may be assembled".

* Delete "A complete round left out of the conditioning cabinet/container over five minutes prior to firing shall be placed at ambient temperature for not less than eight hours and then reconditioned to the applicable temperature."

and substitute the following:

"A complete round left out of the conditioning cabinet/container over five minutes prior to firing shall be placed back in the conditioning cabinet/container for not less than twice the actual time it has been out of conditioning prior to launch."

PAGE 63

4.5.13, paragraph a, Delete the following:

"This time shall be 13.0 $\pm$ 2.0 seconds or 13.5 $\pm$ 2.0 seconds depending on the type of motor used. (see 3.4.1)."

and substitute the following:

"This time shall be 13.5 $\pm$ 2.0 seconds or 13.5 $\pm$ 2.5 seconds depending on the type of motor used. (see 3.4.1)."

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